Baseline Maths Assessment

Year 5

This resource corresponds with the New National Curriculum objectives for Year 4, and has been designed to be used as an aide in assessing pupils' baseline mathematical knowledge at the start of Year 5.

The content addresses each curricular objective outlined in the New National Curriculum in chronological order to give insight into strengths or gaps in each child's mathematical knowledge.

More Assessment resources.

Baseline Maths Assessment – Year 5

Year 4 National Curriculum Objectives:

Number – number and place value

1. Count in multiples of 6, 7, 9, 25 and 1000

2. Find 1000 more or less than a given number

3. Count backwards through zero to include negative numbers

4. Recognise the place value of each digit in a four-digit number (thousands, hundreds, tens, and ones)

5. Order and compare numbers beyond 1000

6. Identify, represent and estimate numbers using different representations

7. Round any number to the nearest 10, 100 or 1000

1, 2, 3, 4, 5, 6 & 7. Solve number and practical problems that involve all of the above and with increasingly large positive numbers

8. Read Roman numerals to 100 (I to C) and know that over time, the numeral system changed to include the concept of zero and place value

Number – addition and subtraction

9, 10, 11 & 12. Add and subtract numbers with up to 4 digits using the formal written methods of columnar addition and subtraction where appropriate

13, 14, 15 & 16. Estimate and use inverse operations to check answers to a calculation

17. Solve addition and subtraction two-step problems in contexts, deciding which operations and methods to use and why

Baseline Maths Assessment - Year 5

Year 4 National Curriculum Objectives:

Number – multiplication and division

18, 19 & 20. Recall multiplication and division facts for multiplication tables up to 12 × 12

21, 22 & 23. Use place value, known and derived facts to multiply and divide mentally, including: multiplying by 0 and 1; dividing by 1; multiplying together three numbers

24. Recognise and use factor pairs and commutativity in mental calculations

25 & 26. Multiply two-digit and three-digit numbers by a one-digit number using formal written layout

27. Solve problems involving multiplying and adding, including using the distributive law to multiply two digit numbers by one digit, integer scaling problems and harder correspondence problems such as n objects are connected to m objects

Number – fractions (including decimals)

28. Recognise and show, using diagrams, families of common equivalent fractions

29. Count up and down in hundredths; recognise that hundredths arise when dividing an object by one hundred and dividing tenths by ten

30. Solve problems involving increasingly harder fractions to calculate quantities, and fractions to divide quantities, including non-unit fractions where the answer is a whole number

31. Add and subtract fractions with the same denominator

33. Recognise and write decimal equivalents of any number of tenths or hundredths

33. Recognise and write decimal equivalents to 1/4 , 1/2 , 3/4

32. Find the effect of dividing a one- or two-digit number by 10 and 100, identifying the value of the digits in the answer as ones, tenths and hundredths

34. Round decimals with one decimal place to the nearest whole number

35. Compare numbers with the same number of decimal places up to two decimal places

36 & 37. Solve simple measure and money problems involving fractions and decimals to two decimal places

Baseline Maths Assessment – Year 5

Year 4 National Curriculum Objectives:

Measurement

38. Convert between different units of measure [for example, kilometre to metre; hour to minute]

39. Measure and calculate the perimeter of a rectilinear figure (including squares) in centimetres and metres

40. Find the area of rectilinear shapes by counting squares

41.Estimate, compare and calculate different measures, including money in pounds and pence

42. Read, write and convert time between analogue and digital 12- and 24-hour clocks

43. Solve problems involving converting from hours to minutes; minutes to seconds; years to months; weeks to days

Geometry – properties of shapes

44. Compare and classify geometric shapes, including quadrilaterals and triangles, based on their properties and sizes

45. Identify acute and obtuse angles and compare and order angles up to two right angles by size

46. Identify lines of symmetry in 2-D shapes presented in different orientations

47. Complete a simple symmetric figure with respect to a specific line of symmetry

Geometry – position and direction

48. Describe positions on a 2-D grid as coordinates in the first quadrant

49. Describe movements between positions as translations of a given unit to the left/right and up/down

50. Plot specified points and draw sides to complete a given polygon

Statistics

51.Interpret and present discrete and continuous data using appropriate graphical methods, including bar charts and time graphs

52. Solve comparison, sum and difference problems using information presented in bar charts, pictograms, tables and other graphs

Baseline Maths Assessment – Year 5

Number and Place Value

1. Complete the next FOUR multiples of 6, 7, 9, 25 and 1,000.

6	12	18	_____	_____	_____	_____
7	14	21	_____	_____	_____	_____
9	18	27	_____	_____	_____	_____
25	50	75	_____	_____	_____	_____
1,000	2,000	3,000	_____	_____	_____	_____

2. Find 1,000 more and less than the given number.

1,000 less	given number	1,000 more
	4,005	
	3,049	
	1,796	
	9,382	
	23,410	

3. Complete the sequence of numbers.

5 4 3 2 [ ] [ ] [ ] [ ] [ ] [ ]

4. Write down the value of the underlined digit.

34<u>6</u> _____

2,5<u>7</u>0 _____

6,<u>3</u>05 _____

<u>9</u>,207 _____

5. Use the symbols > or < to compare these numbers.

2,349 ☐ 1,943 5,688 ☐ 6,588 2,543 ☐ 2,534

6. Write the following numbers in words.

730 _____

2,350 _____

4,067 _____

7. Round the following numbers to the nearest 10.

46 _____ 62 _____ 75 _____

Round the following numbers to the nearest 100.

240 _____ 368 _____ 545 _____

Round the following numbers to the nearest 1,000.

1,350 _____ 2,760 _____ 4,190 _____

8. Write down the number that these Roman numerals represent.

VIII _____ XL _____

XI _____ XC _____

XVII _____ C _____

Addition and Subtraction

9. 245 + 153 =

10. 3,396 + 2,967 =

11. 856 – 672 =

12. 4,642 – 3,584 =

13. 156 + ☐ = 198

14. ☐ + 264 = 552

15. ☐ – 319 = 275

16. 566 – ☐ = 218

17. Jack bought a pencil case for £2.50, a note pad for 86p and a pack of stickers for 55p. He paid with a £5 note. How much change should Jack be given?

Multiplication and Division

18. 9 x 6 = ☐

19. 12 x 8 = ☐

20. 56 ÷ 7 = ☐

21. 48 ÷ 1 = ☐

22. 7 x 1 = ☐

23. 5 x 2 x 6 = ☐

24. Identify the factor pairs of these numbers.

20 = _____

36 = _____

25. 56 x 9 =

26. 472 x 8 =

27. Sally buys 4 drinks at £1.32 each. At the till, she also uses a voucher that gives her 90p off the final bill. How much does Sally end up paying in total?

Fractions (including decimals)

28. Circle the equivalent fraction.

$\frac{1}{2}$ is equivalent to: $\frac{4}{6}$ $\frac{5}{10}$ $\frac{3}{4}$ $\frac{\quad}{4}$ is equivalent to: $\frac{\quad}{8}$ $\frac{\quad}{6}$ $\frac{4}{16}$

29. Complete the sequences.

$\frac{23}{100}$	$\frac{24}{100}$	$\frac{25}{100}$	$\frac{26}{100}$				
0.44	0.45	0.46	0.47				

30. Calculate these fractions of amounts.

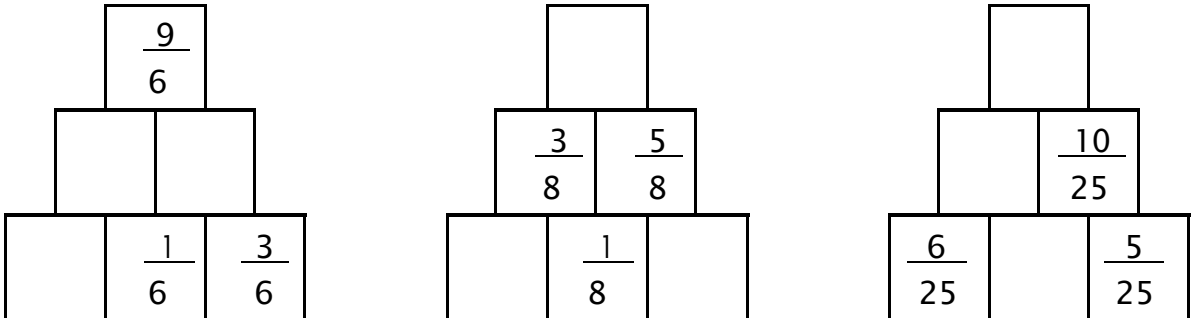
$\frac{1}{5}$ of £30 = ☐

$\frac{1}{8}$ of £56 = ☐

$\frac{2}{3}$ of £27 = ☐

$\frac{3}{4}$ of £48 = ☐

31. Use addition and subtraction to complete the missing fractions.



32. Divide the following numbers by 10 and 100.

67 ÷ 10 = ☐

84 ÷ 10 = ☐

23 ÷ 100 = ☐

59 ÷ 100 = ☐

33. Draw a line to match each fraction to its decimal equivalent.

$\frac{1}{2}$	0.25
$\frac{1}{4}$	0.3
$\frac{3}{4}$	0.5
$\frac{3}{10}$	0.03
$\frac{3}{100}$	0.75

34. Round the following decimals to the nearest whole number.

2.4 ☐ 3.6 ☐ 1.2 ☐ 54.8 ☐

35. Order these numbers from smallest to largest.

5.03	3.53	5.13	3.35	3.05
☐	☐	☐	☐	☐

36. Sunil needs to buy 4 bags of flour. Each bag weighs 2.43kg. What is the total weight of the 4 bags of flour?

37. Ashkay bought 7 bars of chocolate to share amongst his friends. Each bar cost £1.04. How much do the 7 bars cost altogether?

Measurement

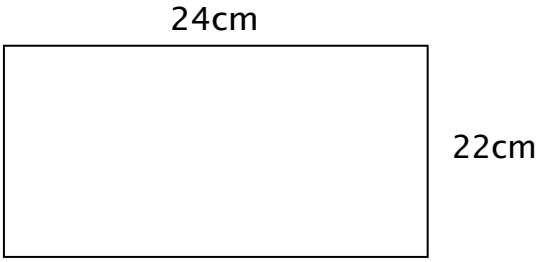
38. Convert the following hours to minutes.

2hrs [mins] 4hrs [mins] 1½hrs [mins]

Convert the following kilometres to metres.

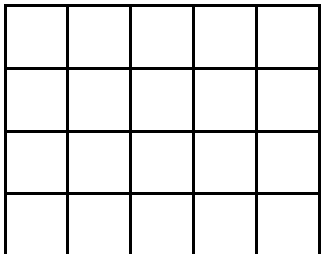
1km [m] 3km [m] 2½km [m]

39. Calculate the perimeter of the following shape.



Perimeter = _____cm

40. Calculate the area of the shape.



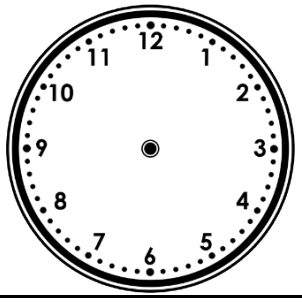
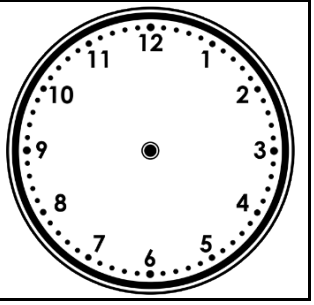
Area = _____ cm²

41. Order the following amounts from largest to smallest.

£1.40 70p £1.04 150p £0.39

[ ] [ ] [ ] [ ] [ ]

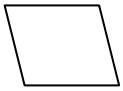
42. Show the given 24hr time on an analogue clock.

14:25		20:50	

43. It's Jenny's birthday in exactly 5 weeks. How many days is that? _____

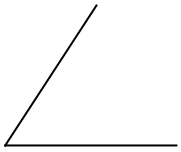
Geometry (properties of shapes)

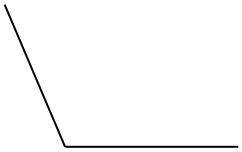
44. Draw a line to connect each shape to the correct name and properties.

parallelogram		a triangle with two equal sides
rhombus		opposite sides are parallel and equal in length
trapezium		1 pair of parallel sides
isosceles triangle		4 equal sides and opposite angles are the same size

45. Add the correct label to the following angles.

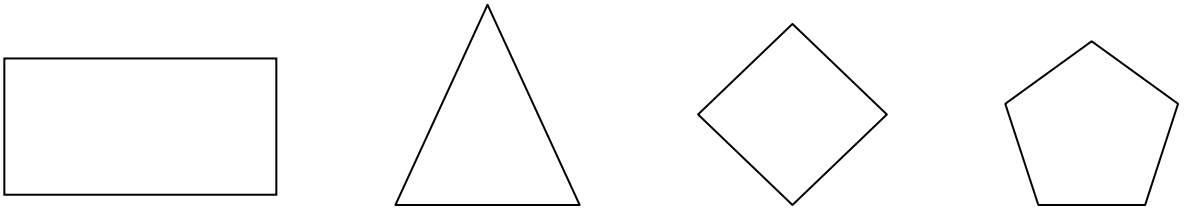
acute	obtuse	right angle



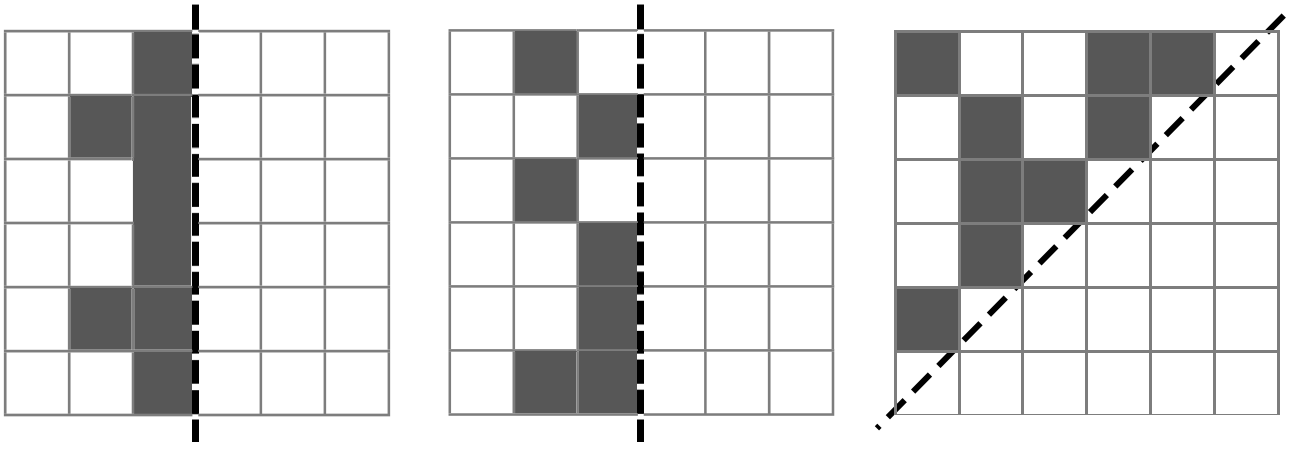


_____ _____ _____

46. Draw lines of symmetry in the following 2D shapes.

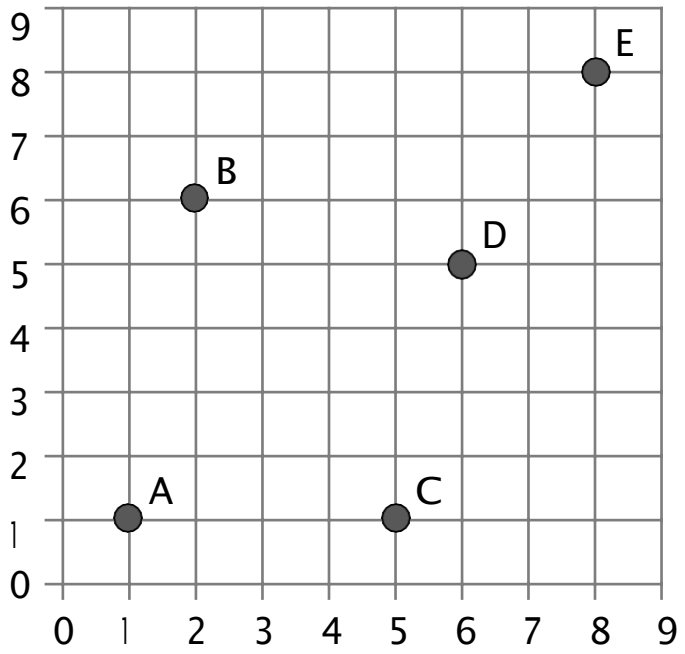


47. Shade in the squares to complete these symmetrical patterns.



Geometry (position and direction)

48. Write down the coordinates for each letter.



A = (,)

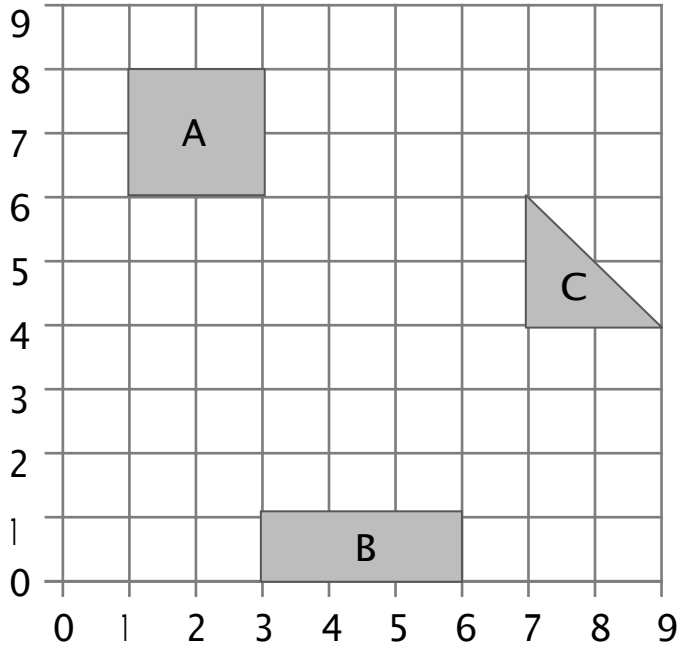
B = (,)

C = (,)

D = (,)

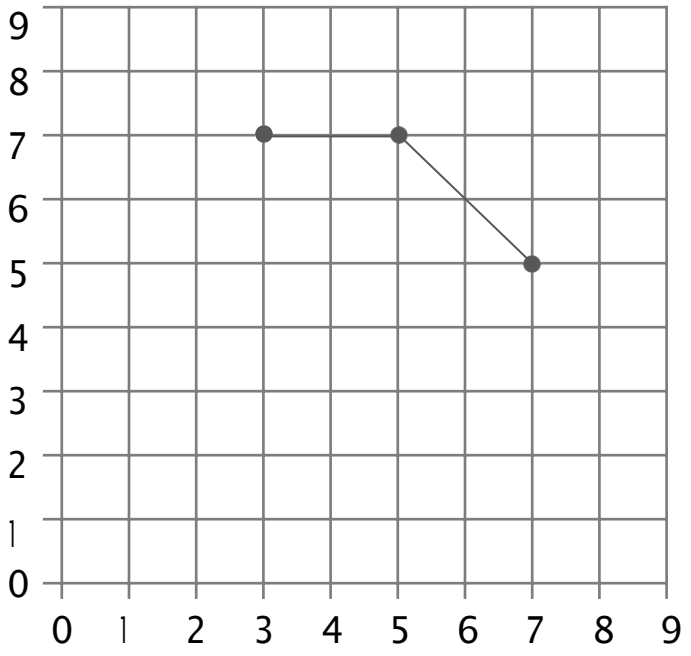
E = (,)

49. Follow the instructions and draw each shape in its new position.



- Translate shape A 3 squares to the right and 4 squares down.
- Translate shape B 3 squares to the left and 4 squares up.
- Translate shape C 2 squares up and 3 squares to the left.

50. Plot the following coordinates on the grid to complete the shape.



(7,3)

(5,1)

(3,1)

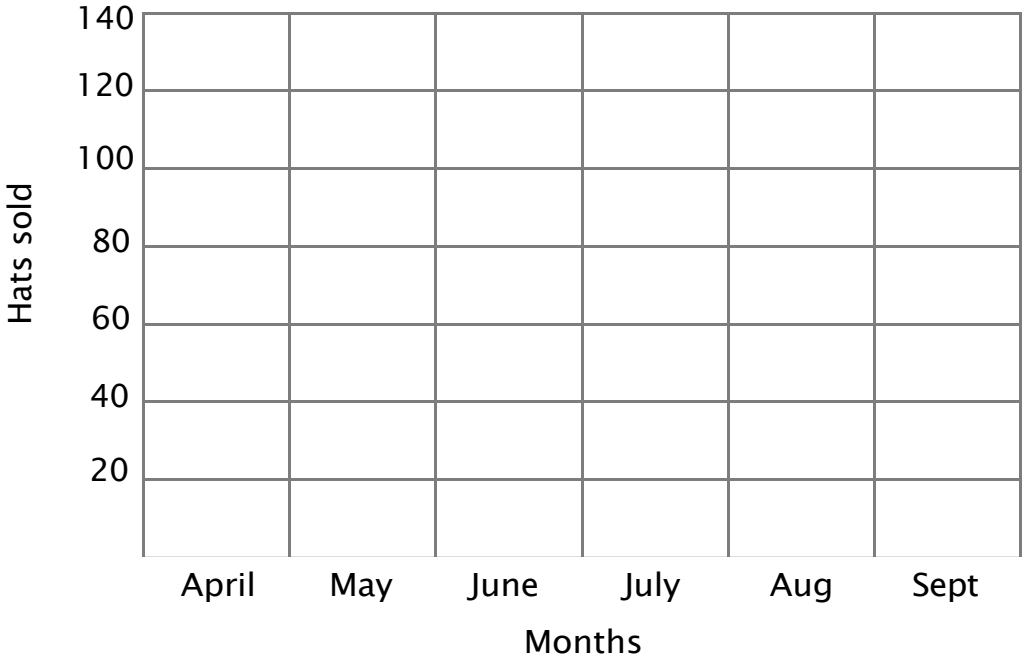
(1,3)

(1,5)

Statistics

51. Complete the graph using the data below.

Month	April	May	June	July	Aug	Sept
Number of hats sold	50	30	110	130	140	110



52. In which month were the greatest number of hats sold? _____

How many more hats were sold in July than in April? _____

How many hats were sold in total from April to September? _____

Baseline Maths Assessment – Year 5

Number and Place Value

1. 24, 30, 36, 42,
   28, 35, 42, 49
   36, 45, 54, 63,
   100, 125, 150, 175
   4,000; 5,000; 6,000; 7,000

2.

1,000 less	given number	1,000 more
3,005	4,005	5,005
2,049	3,049	4,049
796	1,796	2,796
8,382	9,382	10,382
22,410	23,410	24,410

3. 1, 0, -1, -2, -3, -4

4. 6 ones
   7 tens
   3 hundreds
   9 thousands

5. >, <, >

6. seven hundred and thirty
   two thousand, three hundred and fifty
   four thousand and sixty-seven

7. 50, 60, 80
   200, 400, 500
   1,000, 3,000, 4,000

8.

VIII	8	XL	40
XI	11	XC	90
XVII	17	C	100

Addition and Subtraction

9. 398 10. 6,363 11. 184 12. 1,058

Baseline Maths Assessment – Year 5

13. 42 14. 288 15. 594 16. 348

17. £1.09

Multiplication and Division

18. 54 19. 96 20. 8 21. 48 22. 7 23. 60

24. 20 x 1, 10 x 2, 5 x 4
36 x 1, 18 x 2, 12 x 3, 9 x 4, 6 x 6

25. 504

26. 3,776

27. £4.38

Fractions (including decimals)

28. $\frac{1}{2}$ is equivalent to: $\frac{4}{6}$ $\frac{5}{10}$ $\frac{3}{4}$ $\frac{1}{4}$ is equivalent to: $\frac{7}{8}$ $\frac{3}{6}$ $\frac{4}{16}$

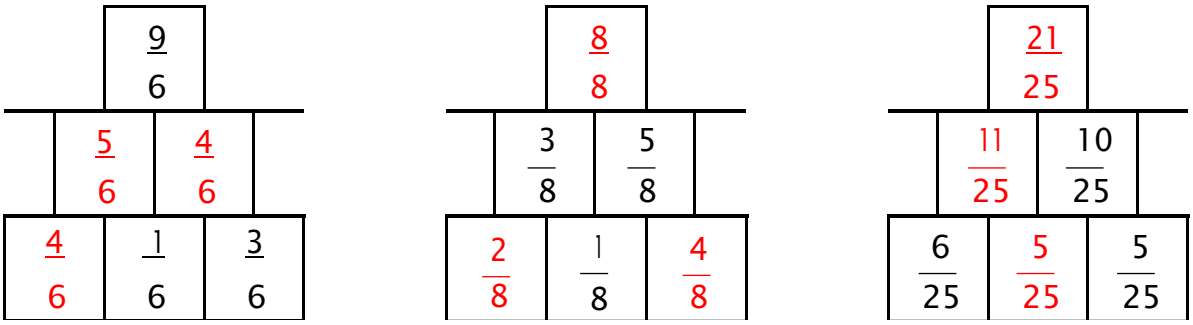
29. $\frac{27}{100}$ $\frac{28}{100}$ $\frac{29}{100}$ $\frac{30}{100}$

0.48 0.49 0.50 0.51

30. $\frac{1}{5}$ of £30 = £6 $\frac{1}{8}$ of £56 = £7

26. $\frac{2}{3}$ of £27 = £18 $\frac{3}{4}$ of £48 = £36

31.



32. 67 ÷ 10 = 6.7 84 ÷ 10 = 8.4

23 ÷ 100 = 0.23 59 ÷ 100 = 0.59

Baseline Maths Assessment – Year 5

33.

$\frac{1}{2}$	0.25
$\frac{1}{4}$	0.3
$\frac{3}{4}$	0.5
$\frac{3}{10}$	0.03
$\frac{3}{100}$	0.75

34. 2, 4, 1, 55

35. 3.05, 3.35, 3.53, 5.03, 5.13

36. 9.72kg

37. £7.28

Measurement

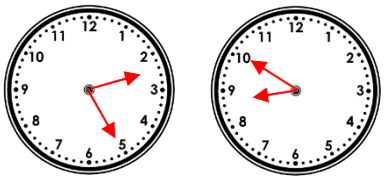
38. 120mins, 240mins, 90mins
1,000m, 3,000m, 2,500m

39. 92cm

40. 20cm²

41. 150p, £1.40, £1.04, 70p, £0.39

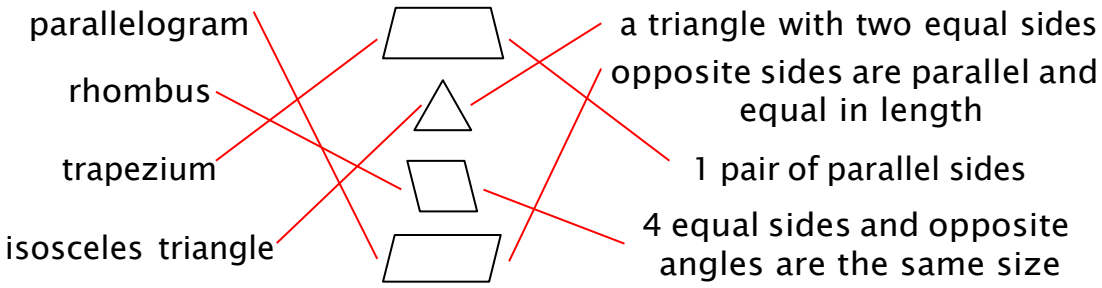
42.



43. 35 days

Geometry (properties of shapes)

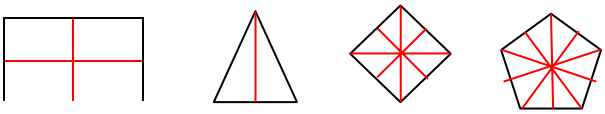
44.



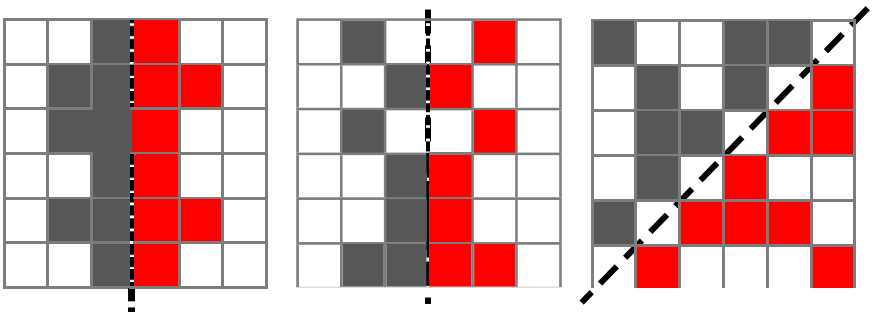
Baseline Maths Assessment – Year 5

45. right angle, acute, obtuse

46.



47.

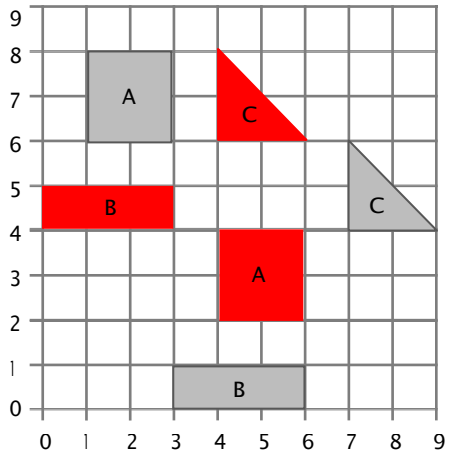


Geometry (position and direction)

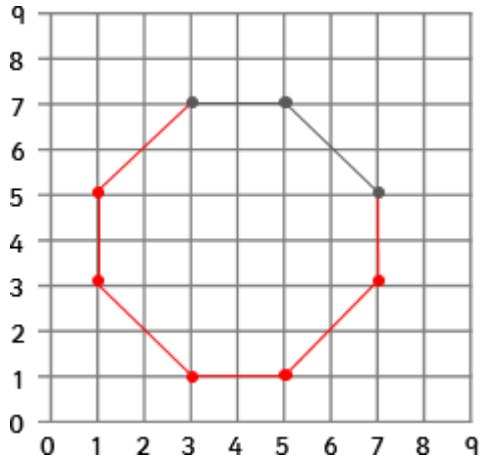
48. A = (1,1), B = (2,6), C = (5,1), D = (6,5), E = (8,8)

49.

50.

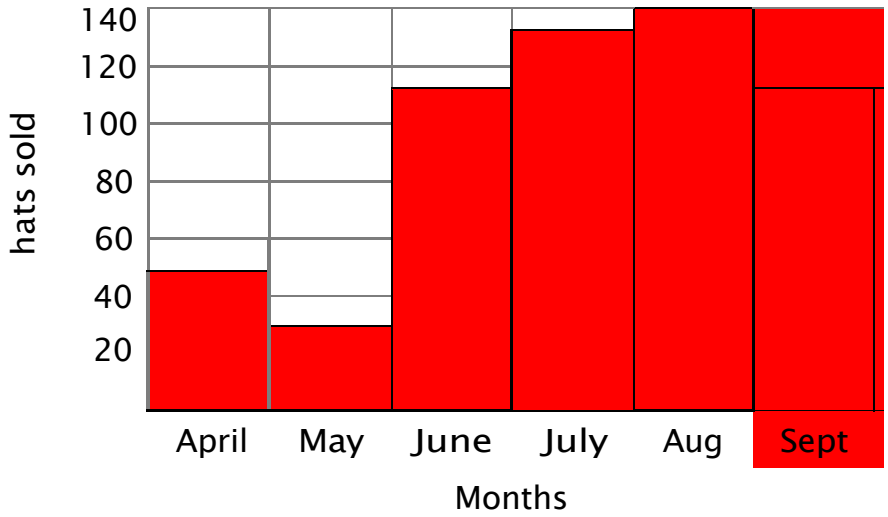


50.



Statistics

51.



52. August, 80, 570